

Run: overnight-cl250

Log-log loss curves

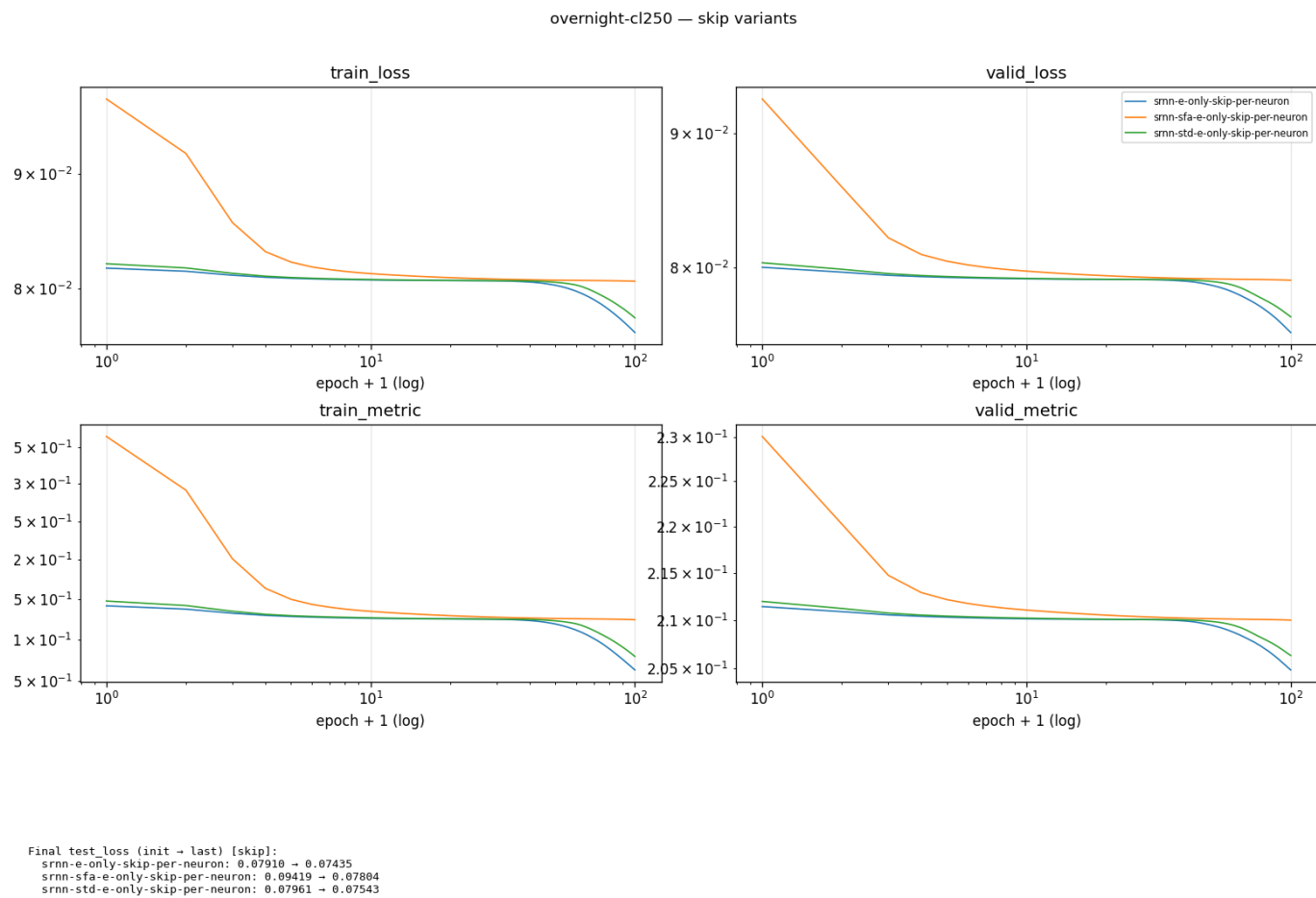


Figure 1: skip variants — log-log loss/metric

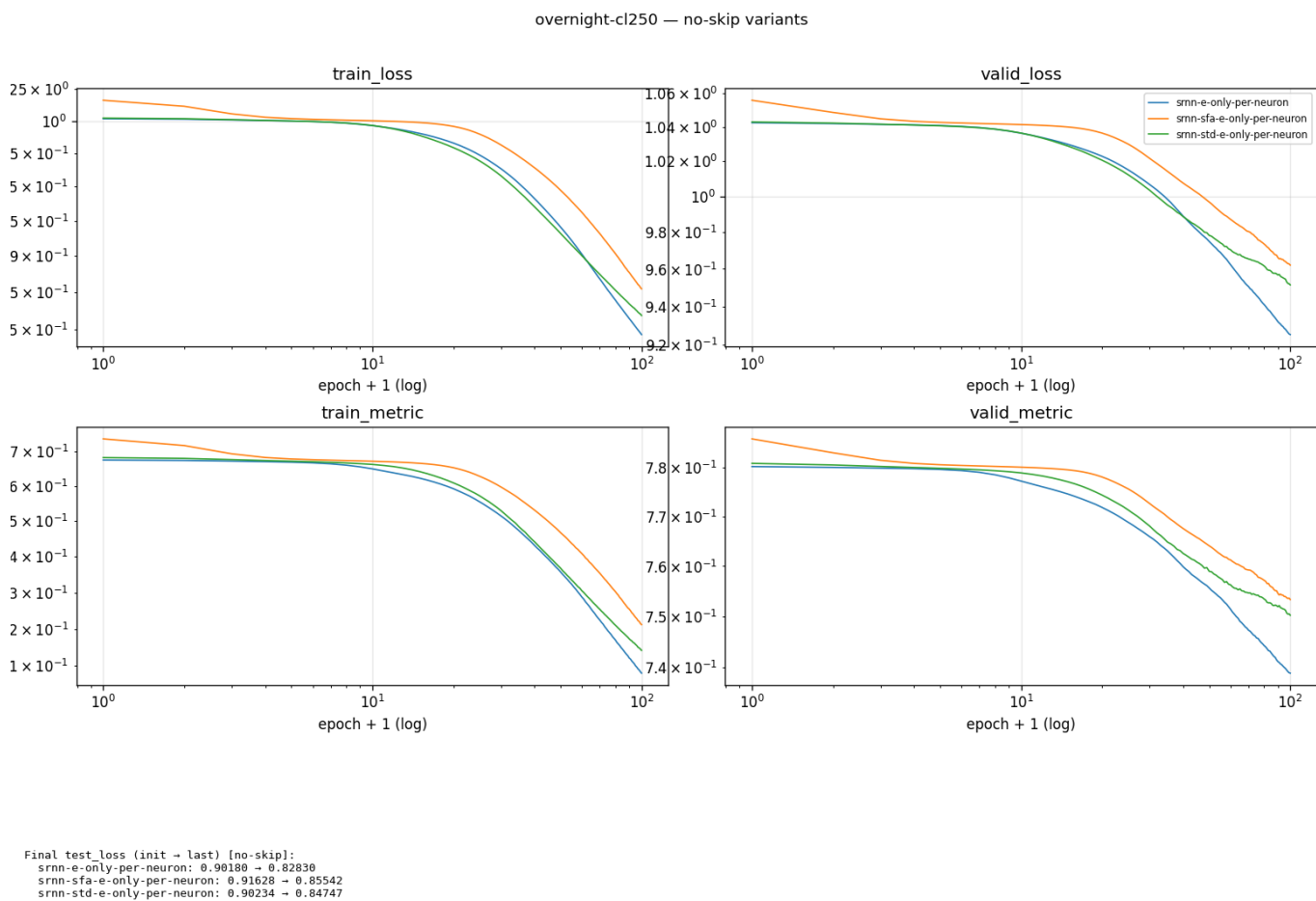


Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

overnight-cl250 — effective time constants (srnn-e-only-per-neuron)

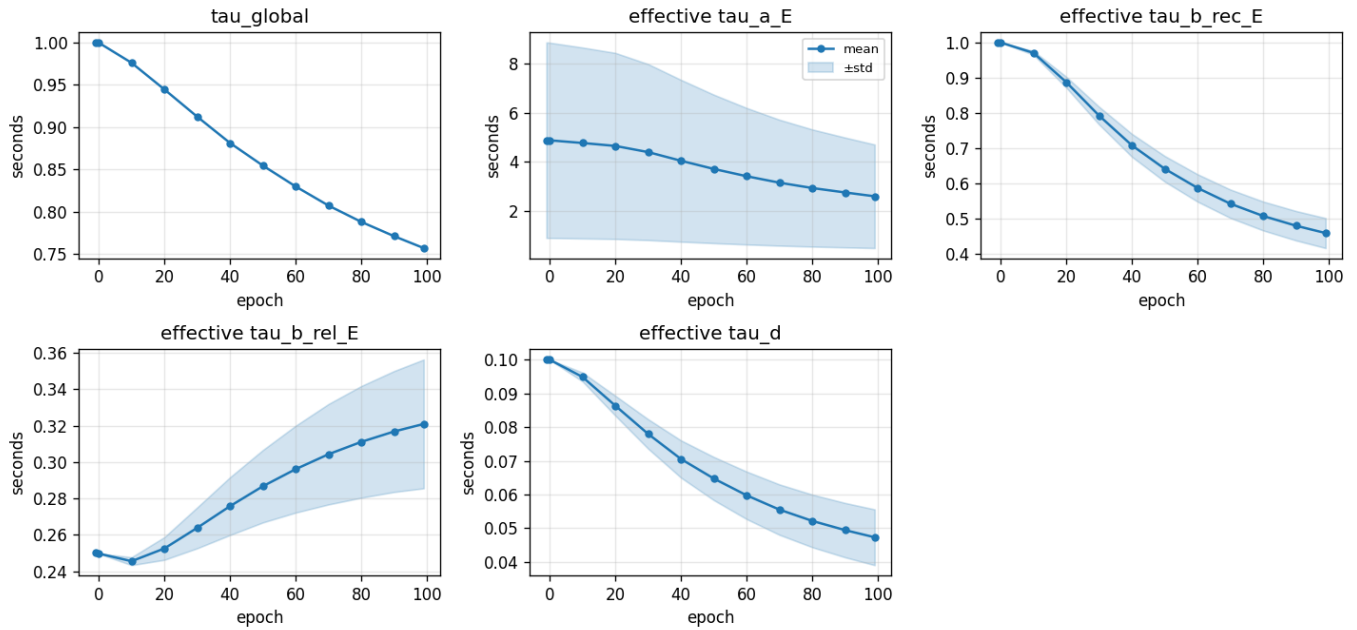
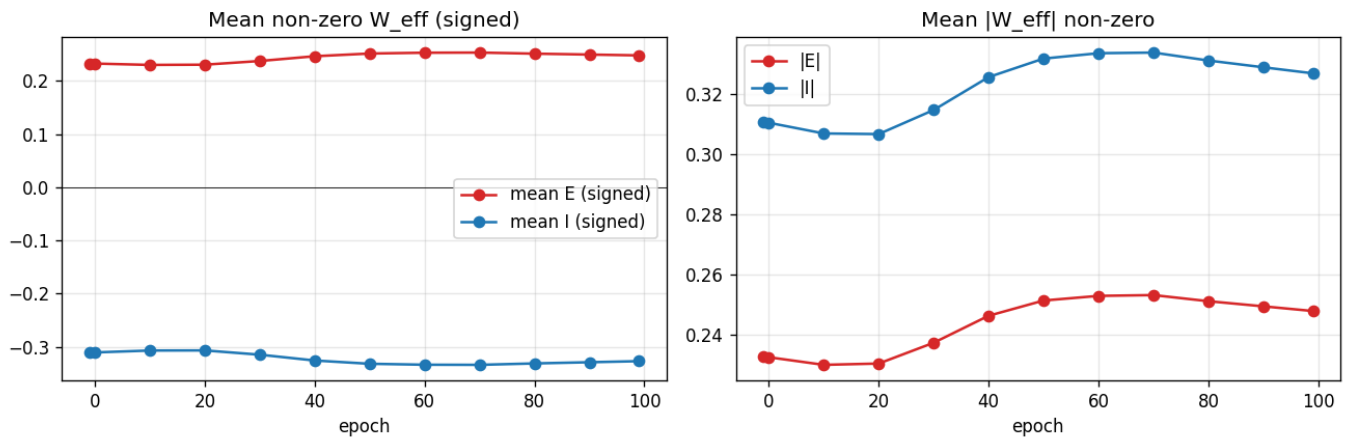


Figure 3: Tau evolution

overnight-cl250 — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

overnight-cl250 - variant srnn-e-only-per-neuron (k=0)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.75694	-	-	scalar
tau_d (s)	+0.1	+0.047285	0	0.00829	300
tau_a_E (s)	+4.8777	+2.5842	3.996	2.116	450
tau_b_rec_E (s)	+1	+0.45844	0	0.04265	150
tau_b_rel_E (s)	+0.25	+0.32092	0	0.03539	150
W_eff (E src, signed)	+0.23267	+0.24787	0.07641	0.08534	15048
W_eff (I src, signed)	-0.31062	-0.32687	0.07724	0.08695	14977
W_eff (E src)	+0.23267	+0.24787	0.07641	0.08534	15048
W_eff (I src)	+0.31062	+0.32687	0.07724	0.08695	14977
W_in_eff (input neurons)	-0.0003386	+0.0013842	0.1012	0.1342	6675
c_E (SFA coupling)	+0.05	+0.05476	0	0.006898	450
c_0_E (SFA offset)	+0	-0.27431	0	0.09773	450
a_0 (threshold)	+0.35	+0.16777	0	0.08514	300
readout_weight	-2.8854e-05	-0.0057749	0.06604	0.1456	6675
readout_bias	+0	+0.0044741	0	0.1548	89
ic.ic (per-variant init)	+0.098875	-0.00036654	0.3842	0.5642	900

srnn-e-only-skip-per-neuron

overnight-cl250 — effective time constants (srnn-e-only-skip-per-neuron)

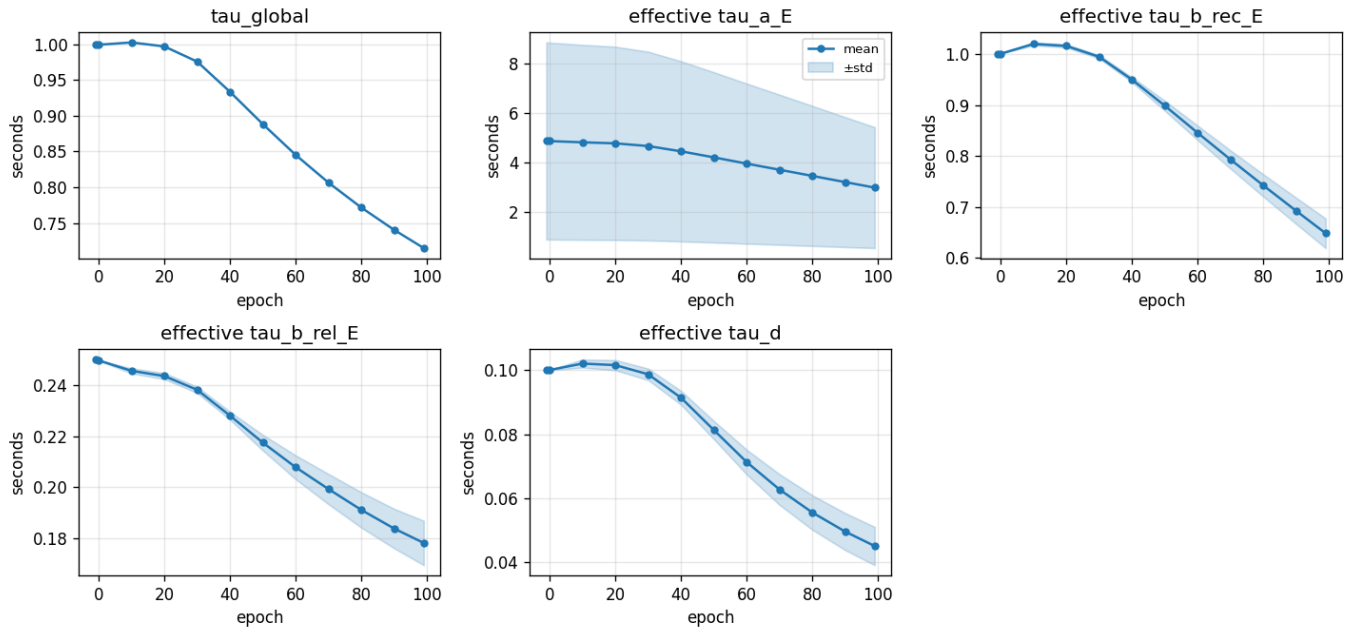
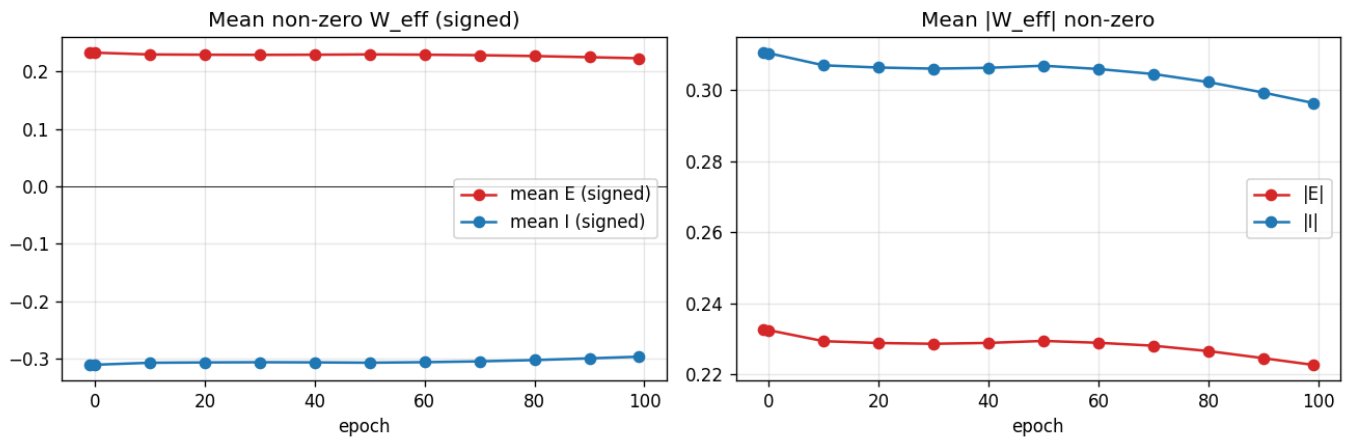


Figure 5: Tau evolution

overnight-cl250 — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

overnight-cl250 - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.71474	-	-	scalar
tau_d (s)	+0.1	+0.045103	0	0.00597	300
tau_a_E (s)	+4.8777	+2.9855	3.996	2.453	450
tau_b_rec_E (s)	+1	+0.64798	0	0.02907	150
tau_b_rel_E (s)	+0.25	+0.17806	0	0.008715	150
W_eff (E src, signed)	+0.23267	+0.22277	0.07641	0.07386	15048
W_eff (I src, signed)	-0.31062	-0.29646	0.07724	0.07555	14977
W_eff (E src)	+0.23267	+0.22277	0.07641	0.07386	15048
W_eff (I src)	+0.31062	+0.29646	0.07724	0.07555	14977
W_in_eff (input neurons)	-0.00084237	+0.001791	0.1008	0.0739	6675
c_E (SFA coupling)	+0.05	+0.052935	0	0.003242	450
c_0_E (SFA offset)	+0	-0.0012899	0	0.05942	450
a_0 (threshold)	+0.35	+0.28407	0	0.07765	300
readout_weight	+0.00078219	-0.013388	0.06654	0.07669	6675
readout_bias	+0	+0.075552	0	0.08734	89
ic.ic (per-variant init)	+0.098961	+0.064123	0.387	0.4347	900

srnn-sfa-e-only-per-neuron

overnight-cl250 — effective time constants (srnn-sfa-e-only-per-neuron)

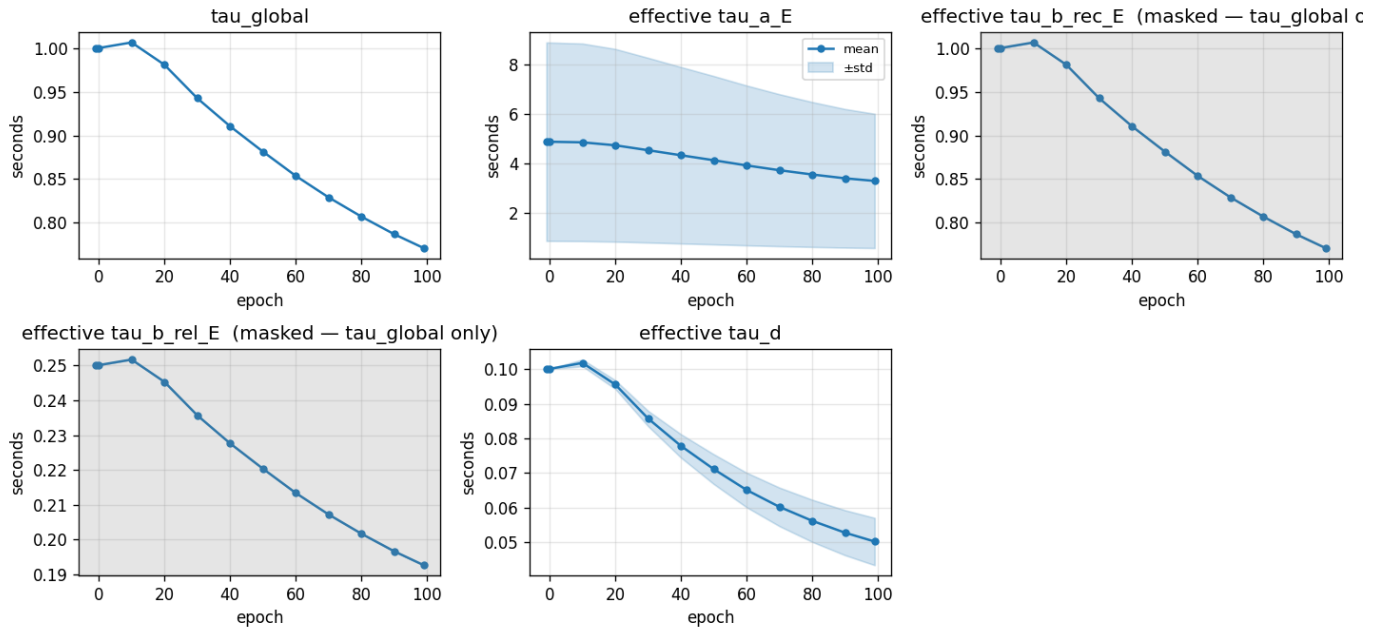
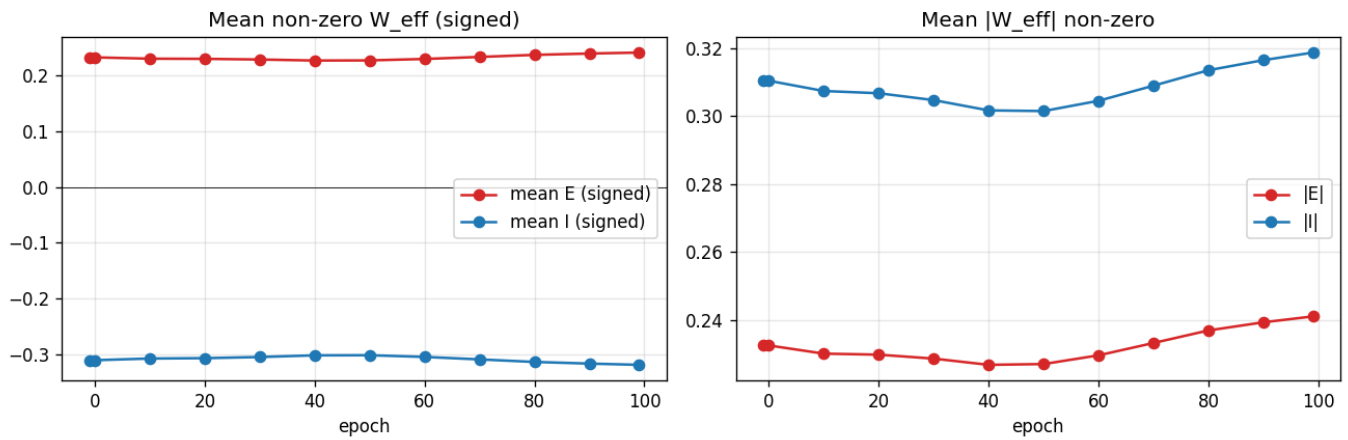


Figure 7: Tau evolution

overnight-cl250 — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

overnight-cl250 - variant srnn-sfa-e-only-per-neuron (k=2)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.77065	-	-	scalar
tau_d (s)	+0.1	+0.050201	0	0.006832	300
tau_a_E (s)	+4.8777	+3.2983	3.996	2.703	450
tau_b_rec_E (s)	+1	+0.77065	0	0	150
tau_b_rel_E (s)	+0.25	+0.19266	0	0	150
W_eff (E src, signed)	+0.23267	+0.24106	0.07641	0.08131	15048
W_eff (I src, signed)	-0.31062	-0.31873	0.07724	0.08281	14977
W_eff (E src)	+0.23267	+0.24106	0.07641	0.08131	15048
W_eff (I src)	+0.31062	+0.31873	0.07724	0.08281	14977
W_in_eff (input neurons)	-0.00015382	-0.0020851	0.09934	0.1374	6675
c_E (SFA coupling)	+0.05	+0.051482	0	0.003646	450
c_0_E (SFA offset)	+0	-0.079732	0	0.06781	450
a_0 (threshold)	+0.35	+0.34909	0	0.05572	300
readout_weight	-0.00098972	+0.0052418	0.06678	0.1206	6675
readout_bias	+0	-0.026226	0	0.06046	89
ic.ic (per-variant init)	+0.13078	+0.065655	0.4172	0.4345	900

srnn-sfa-e-only-skip-per-neuron

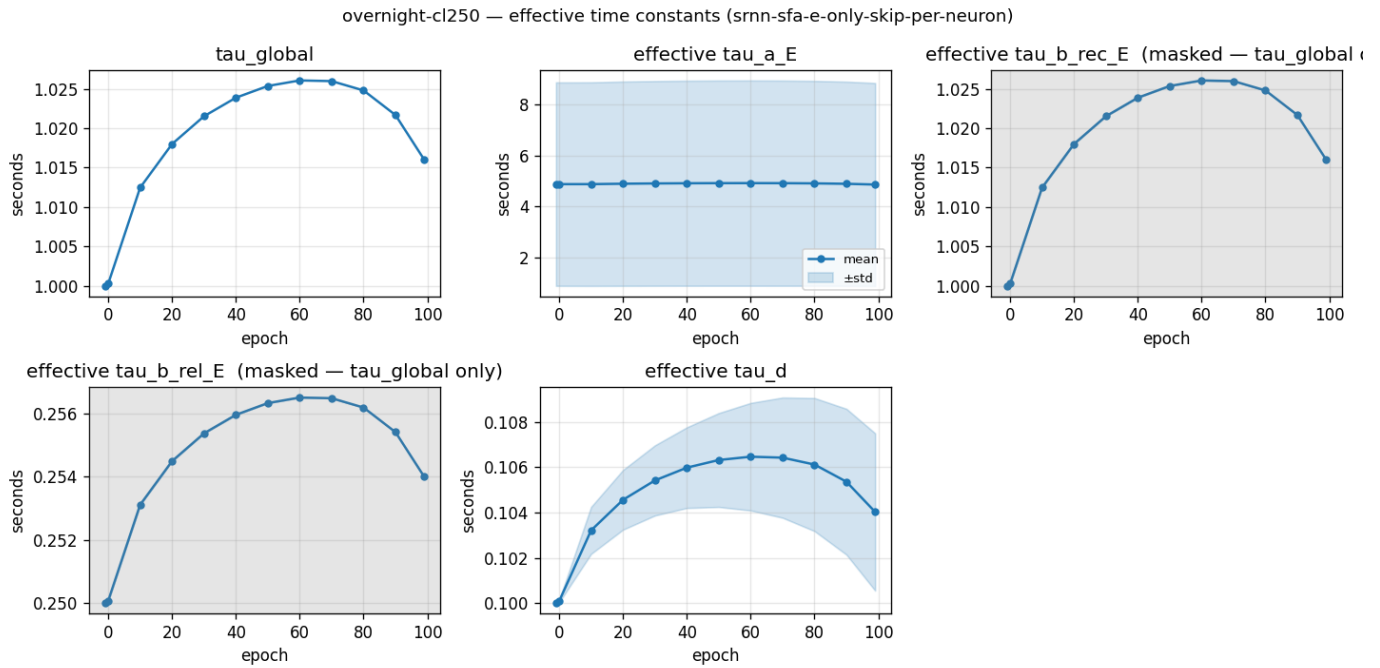
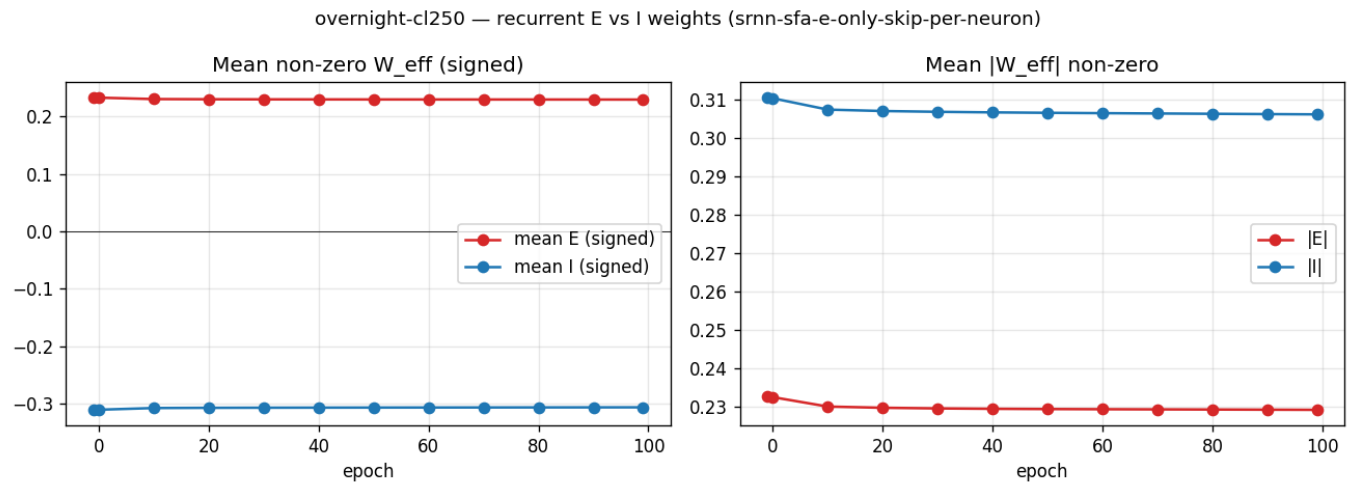


Figure 9: Tau evolution

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

overnight-cl250 - variant srnn-sfa-e-only-skip-per-neuron (k=3)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+1.016	-	-	scalar
tau_d (s)	+0.1	+0.10403	0	0.003474	300
tau_a_E (s)	+4.8777	+4.8605	3.996	3.986	450
tau_b_rec_E (s)	+1	+1.016	0	0	150
tau_b_rel_E (s)	+0.25	+0.254	0	0	150
W_eff (E src, signed)	+0.23267	+0.22921	0.07641	0.07535	15048
W_eff (I src, signed)	-0.31062	-0.30625	0.07724	0.07637	14977
W_eff (E src)	+0.23267	+0.22921	0.07641	0.07535	15048
W_eff (I src)	+0.31062	+0.30625	0.07724	0.07637	14977
W_in_eff (input neurons)	-0.0011621	-0.0014667	0.09962	0.09388	6675
c_E (SFA coupling)	+0.05	+0.051699	0	0.001434	450
c_0_E (SFA offset)	+0	+0.036936	0	0.02567	450
a_0 (threshold)	+0.35	+0.37426	0	0.03578	300
readout_weight	+0.00016801	-0.0010738	0.0673	0.06088	6675
readout_bias	+0	+0.0022042	0	0.003744	89
ic.ic (per-variant init)	+0.09231	+0.11729	0.5294	0.4143	900

srnn-std-e-only-per-neuron

overnight-cl250 — effective time constants (srnn-std-e-only-per-neuron)

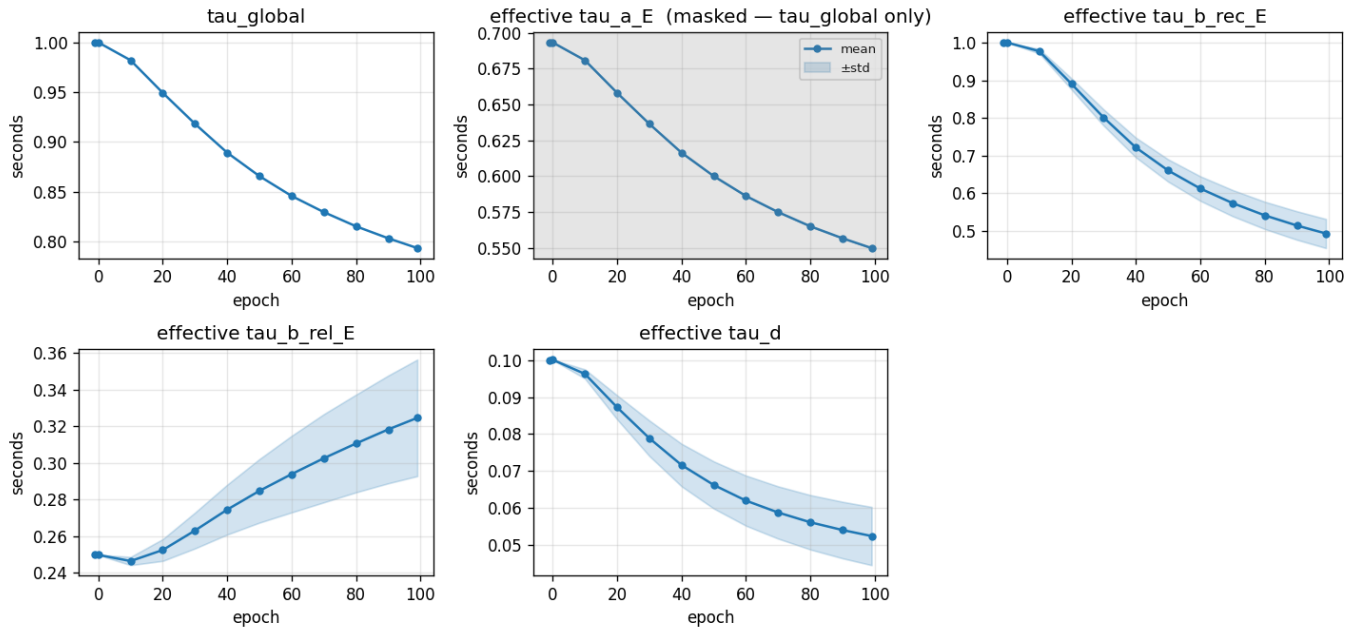
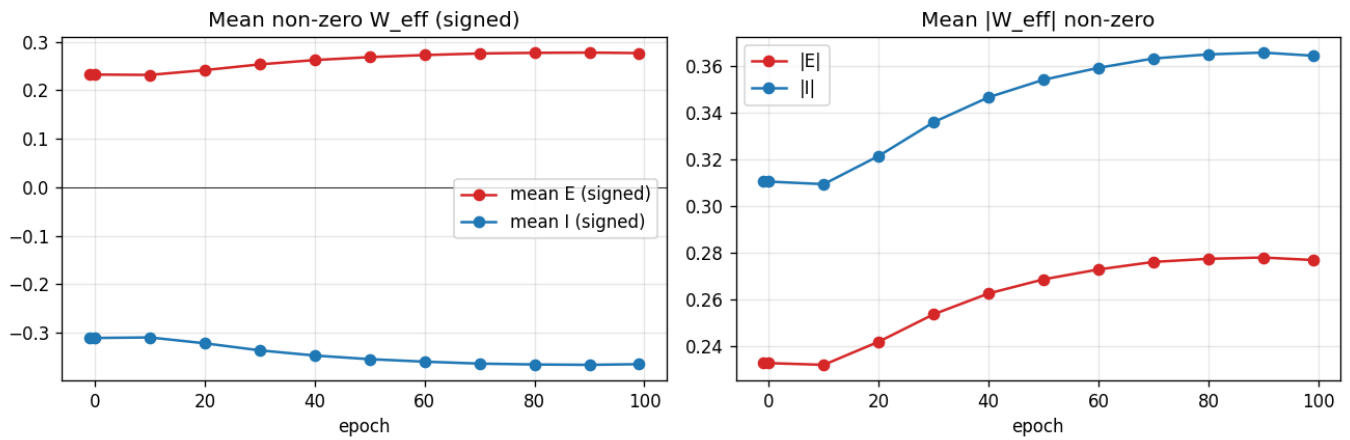


Figure 11: Tau evolution

overnight-cl250 — recurrent E vs I weights (srnn-std-e-only-per-neuron)

Figure 12: W_{EI} evolution

srnn-std-e-only-per-neuron — parameter table

overnight-cl250 - variant srnn-std-e-only-per-neuron (k=4)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.79317	-	-	scalar
tau_d (s)	+0.1	+0.052401	0	0.007879	300
tau_a_E (s)	+0.69315	+0.54978	0	0	450
tau_b_rec_E (s)	+1	+0.49281	0	0.03874	150
tau_b_rel_E (s)	+0.25	+0.32451	0	0.03178	150
W_eff (E src, signed)	+0.23267	+0.27677	0.07641	0.09482	15048
W_eff (I src, signed)	-0.31062	-0.36449	0.07724	0.09683	14977
W_eff (E src)	+0.23267	+0.27677	0.07641	0.09482	15048
W_eff (I src)	+0.31062	+0.36449	0.07724	0.09683	14977
W_in_eff (input neurons)	+0.00019944	-0.00091659	0.09994	0.1406	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.22171	0	0.08034	300
readout_weight	+0.00074661	+0.0096444	0.06657	0.1413	6675
readout_bias	+0	-0.022083	0	0.1547	89
ic.ic (per-variant init)	+0.087739	+0.028283	0.3868	0.4661	900

srnn-std-e-only-skip-per-neuron

overnight-cl250 — effective time constants (srnn-std-e-only-skip-per-neuron)

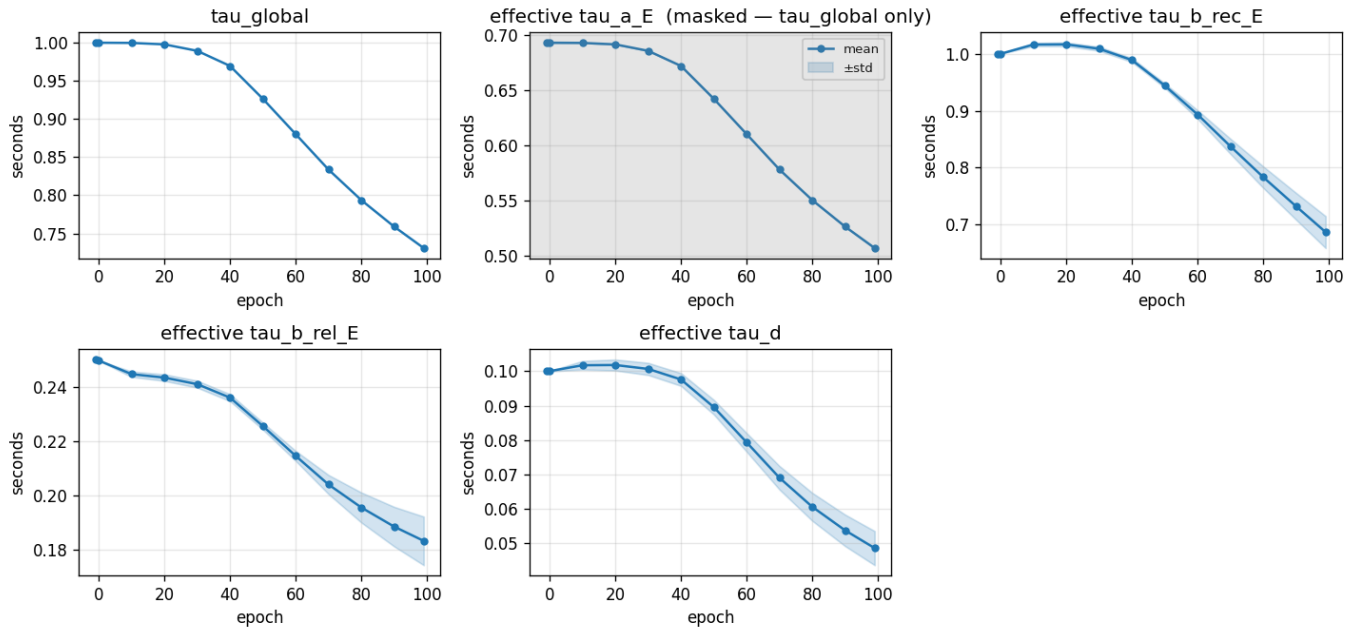


Figure 13: Tau evolution

overnight-cl250 — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

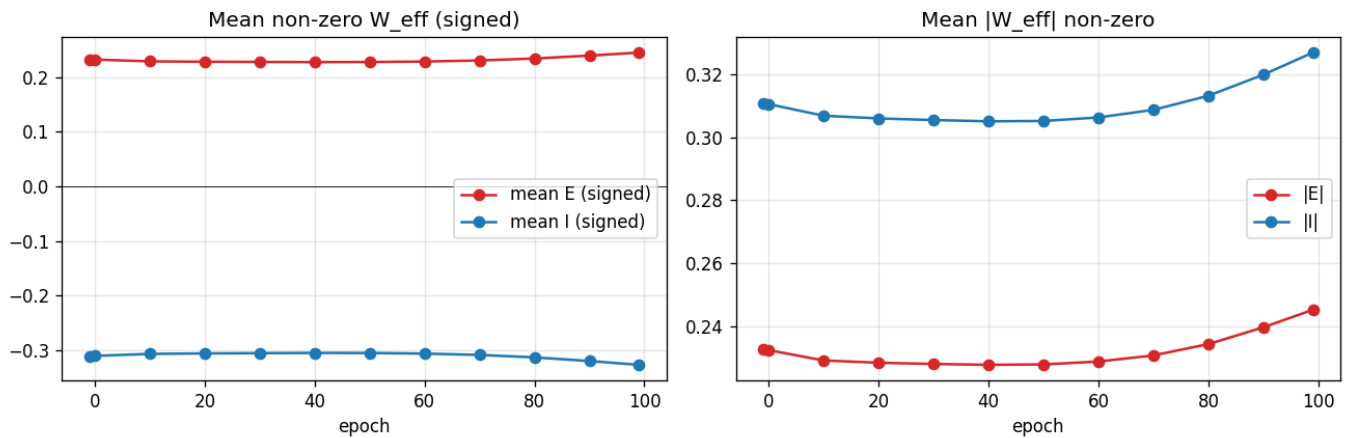


Figure 14: W_EI evolution

srnn-std-e-only-skip-per-neuron — parameter table

overnight-cl250 - variant srnn-std-e-only-skip-per-neuron (k=5)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.73075	-	-	scalar
tau_d (s)	+0.1	+0.048476	0	0.004967	300
tau_a_E (s)	+0.69315	+0.50652	0	0	450
tau_b_rec_E (s)	+1	+0.68599	0	0.02816	150
tau_b_rel_E (s)	+0.25	+0.18328	0	0.008936	150
W_eff (E src, signed)	+0.23267	+0.24529	0.07641	0.08121	15048
W_eff (I src, signed)	-0.31062	-0.3268	0.07724	0.08245	14977
W_eff (E src)	+0.23267	+0.24529	0.07641	0.08121	15048
W_eff (I src)	+0.31062	+0.3268	0.07724	0.08245	14977
W_in_eff (input neurons)	+0.00036119	-0.00069182	0.09956	0.08771	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.3437	0	0.05943	300
readout_weight	+0.00055372	-0.0083536	0.06572	0.06826	6675
readout_bias	+0	+0.024059	0	0.04786	89
ic.ic (per-variant init)	+0.089141	+0.07427	0.3857	0.4266	900